

Lecture 14

Statistical Significance and an Introduction to Probability

Review

1 What is a sampling distribution?

The distribution of a statistic over all possible samples of the same size n .

2 What is the standard error?

The standard deviation of a sampling distribution; for $\bar{x}$ it is $SE = \sigma / \sqrt{n}$.

3 What is the margin of error?

How far an estimate is expected to fall from the parameter: $2 \times SE$.

4 What is a linear transformation?

$y = ax + b$: it rescales and shifts the center and spread, but never changes the shape.

What you should know: the normal distribution

The two parameters

$x \sim N(\mu, \sigma)$: μ locates the center, σ sets the spread.

The empirical rule

68%, 95% and 99.7% within 1, 2 and 3 standard deviations (bell-shaped only).

The z-score

$z = (x - \bar{x}) / s$ counts standard deviations from the mean; $|z| \geq 2$ flags an outlier.

Data: student heights

A survey of 379 college students recorded each student's height and gender.

X = the height of a student, in inches

HEIGHT	GENDER
56	Female
57	Female
58	Female
60	Female
60	Female
...	...

n = 379 students (the file lists the women first, then the men, each sorted by height)

Statistical significance

Someone hands you 20 students and says they are a random sample of all 379.

$\bar{x} = 70.33$ in $\mu = 67.1$ in $SE = 0.92$ in

Is this result statistically significant?

$z = (70.33 - 67.1) / 0.92 \approx 3.5$: far outside the margin of error, so yes

Statistically significant: this was not a random sample of all students (they were all men)

Why probability?

Wednesday: about 95% of samples land within 2 SE of μ .

Today: a sample that far out would be very unlikely to happen by chance.

We only ever take one sample. So what do "95%" and "unlikely" actually mean?

**They describe what would happen if we repeated the random sampling many, many times
Statements about the long run of a chance process are statements of probability**

Quantifying uncertainty

randomization

we use chance on purpose - random sampling, random assignment - to avoid bias

the price: uncertainty

by chance, the statistics from a sample are not exactly the parameters of the population

the payoff

chance-based results follow the laws of probability, so we can measure how far off we may be

The language of probability

random trial

a process or experiment with well-defined possible outcomes; it is random because more than one outcome is possible

sample space S

the collection of all possible outcomes: for one roll of a die, $S = \{1, 2, 3, 4, 5, 6\}$

event

any possible outcome of a random trial, written with a capital letter: $A = \text{roll an even number} = \{2, 4, 6\}$

probability $P(A)$

a number between 0 and 1 that measures how likely event A is: the proportion of times A occurs in a very long series of repetitions

Equally likely outcomes

Roll a fair six-sided die: all six outcomes are equally likely.

$P(A) = (\text{number of ways } A \text{ can happen}) / (\text{total number of outcomes})$

(a) What is $P(\text{even})$?

(b) What is $P(\text{not a six})$?

(a) $3 / 6 = 0.5$

(b) $5 / 6 \approx 0.83$

Example: probabilities from real data

Pick one of the 379 students from the heights data at random.

	At least 70 in	Under 70 in	Total
Female	19	243	262
Male	87	30	117
Total	106	273	379

(a) What is $P(\text{the student is at least 70 in})$?

(b) What is $P(\text{the student is female})$?

(a) $106 / 379 \approx 0.28$

(b) $262 / 379 \approx 0.69$

Where do probabilities come from?

Flip a fair coin once. Intuition says $P(\text{heads}) = 1/2$.

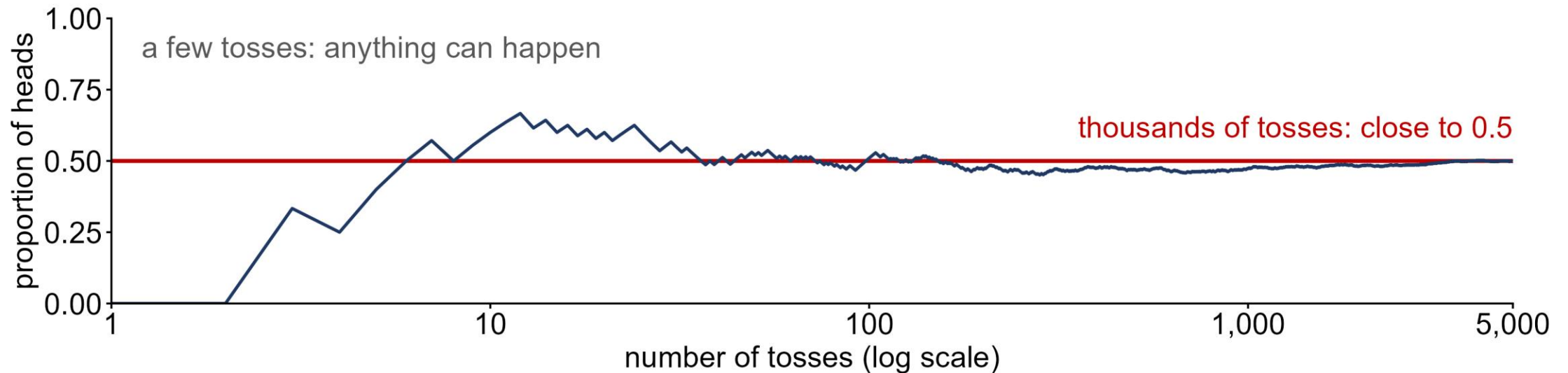
You flip it 4 times and get 3 heads: 75%. Was $1/2$ wrong?

No: over a few flips the proportion is unpredictable; over many it is very predictable
Probability: the proportion of times an event occurs in a very long series of repetitions

The law of large numbers

law of large numbers

as the number of independent trials n increases, the relative frequency of an event, (number of times A occurs) / n , gets closer and closer to $P(A)$



Independent trials: the outcome of one trial does not affect the outcome of any other

Jacob Bernoulli proved it (published 1713); Karl Pearson tossed a coin 24,000 times: 12,012 heads = 0.5005

The rules of probability

Rule 1: $0 \leq P(A) \leq 1$

the probability of an event must be a value between 0 and 1

Rule 2: $\sum P(A_i) = 1$

the probabilities of all the outcomes in the sample space must add to 1

Rule 3: $P(A \cup B) = P(A) + P(B)$

the addition rule for DISJOINT events: A and B have no outcomes in common

Rule 4: $P(A') = 1 - P(A)$

the complement rule: A' ("A-not") is every outcome in S that is not in A

Combining two events

union $A \cup B$

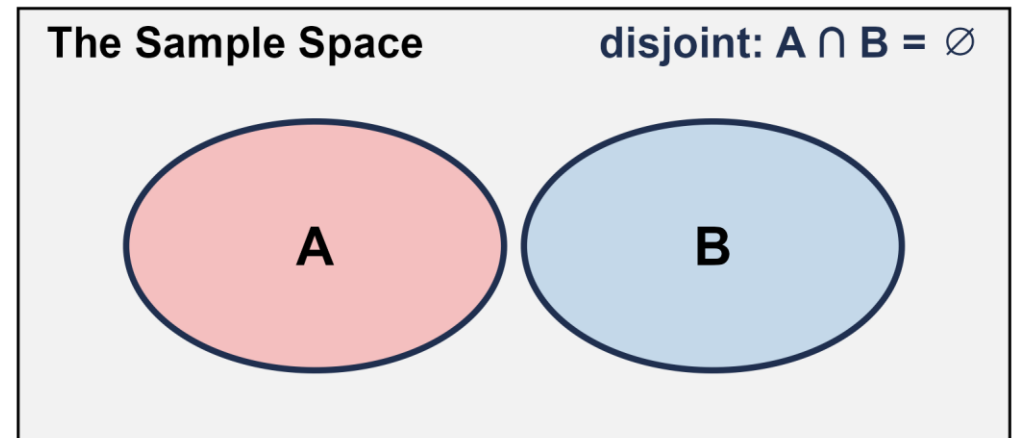
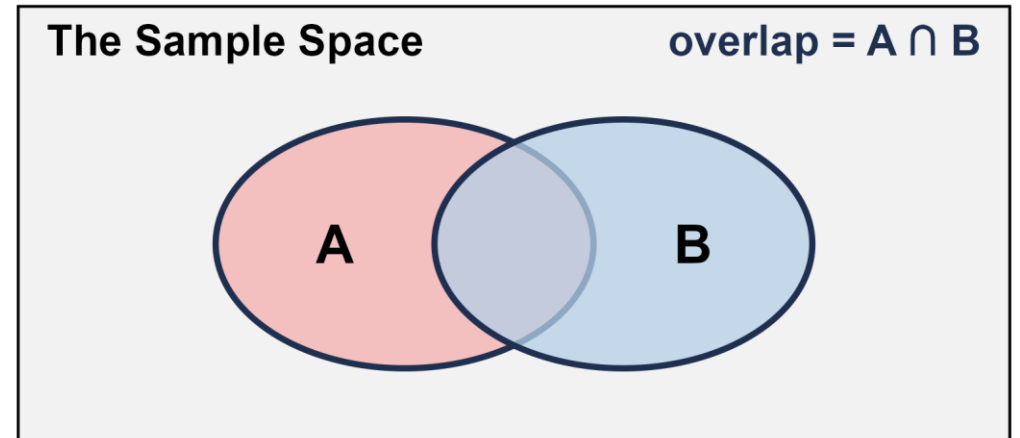
outcomes in A or in B (or both); key word "or"

intersection $A \cap B$

outcomes A and B have in common; key word "and"

disjoint (mutually exclusive)

A and B have no outcomes in common: $A \cap B = \emptyset$



Two more rules

general addition rule

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$: subtract the overlap so it is not counted twice

multiplication rule

if A and B are independent (one occurring does not change the probability of the other): $P(A \cap B) = P(A) \cdot P(B)$

disjoint $\neq$ independent

disjoint events cannot happen together, so knowing one happened tells you the other did not

Example: one die

Roll a fair six-sided die once.

$S = \{1, 2, 3, 4, 5, 6\}$, each outcome has probability $1/6$

- (a) What is $P(\text{not a six})$?
- (b) What is $P(1 \cup 6)$?
- (c) What is $P(\text{even} \cup \text{greater than } 3)$?

- (a) $1 - 1/6 = 5/6$ (rule 4)
- (b) $1/6 + 1/6 = 1/3$ (rule 3: a 1 and a 6 are disjoint)
- (c) $3/6 + 3/6 - 2/6 = 4/6$ (the 4 and the 6 are in both)

Example: two dice

Roll a fair die twice. The two rolls are independent.

$P(\text{six}) = 1/6$ on each roll

(a) What is $P(\text{six on both rolls})$?

(b) What is $P(\text{at least one six})$?

(a) $(1/6)(1/6) = 1/36$ (multiplication rule)

(b) $1 - P(\text{no sixes}) = 1 - (5/6)(5/6) = 11/36 \approx 0.31$

Example: unions and intersections

Pick one of the 379 students at random. F = female, T = at least 70 in.

	At least 70 in	Under 70 in	Total
Female	19	243	262
Male	87	30	117
Total	106	273	379

(a) What is $P(F \cap T)$?

(b) What is $P(F \cup T)$?

(a) $19 / 379 \approx 0.05$

(b) $262/379 + 106/379 - 19/379 = 349/379 \approx 0.92$

Are F and T independent?

Pick one of the 379 students at random. F = female, T = at least 70 in.

	At least 70 in	Under 70 in	Total
Female	19	243	262
Male	87	30	117
Total	106	273	379

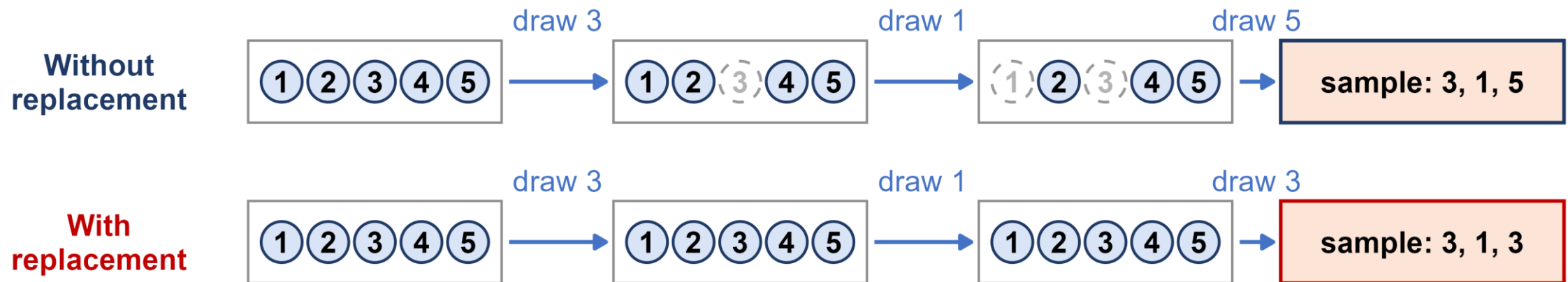
Does knowing a student is female change the chance they are at least 70 in?

Yes: $106/379 \approx 28\%$ of all students, but only $19/262 \approx 7\%$ of the women

Not independent: $P(F) \cdot P(T) \approx 0.69 \cdot 0.28 \approx 0.19$, but $P(F \cap T) \approx 0.05$

With or without replacement?

Week 4: drawing a sample of 3 balls from a bag of 5, two ways.



Are the draws independent?

With replacement: yes. The bag is the same on every draw, so $P(A \cap B) = P(A) \cdot P(B)$

Without replacement: no. Each draw changes what is left, so the next probability depends on the last draw

Example: drawing two aces

A standard deck has 52 cards, 4 of them aces. Draw two cards.

$P(\text{ace on the first draw}) = 4/52$

- (a) What is $P(\text{two aces})$ if the first card is put back (with replacement)?**
- (b) What is $P(\text{two aces})$ if it is not put back (without replacement)?**

- (a) $(4/52)(4/52) = 1/169 \approx 0.0059$: the draws are independent**
- (b) $(4/52)(3/51) = 1/221 \approx 0.0045$: only 3 aces are left among 51 cards**

Random variables

random variable

the outcome of a random trial recorded as a number, written with a capital letter such as X

discrete

its possible values can be listed: the number of heads in 3 tosses, $X = 0, 1, 2, 3$

continuous

it can take any value in an interval: a student's height

Example: three coin tosses

Toss a fair coin 3 times: $2 \times 2 \times 2 = 8$ equally likely outcomes.

X = the number of heads

(a) List the sample space S.

(b) How many outcomes give each value of X?

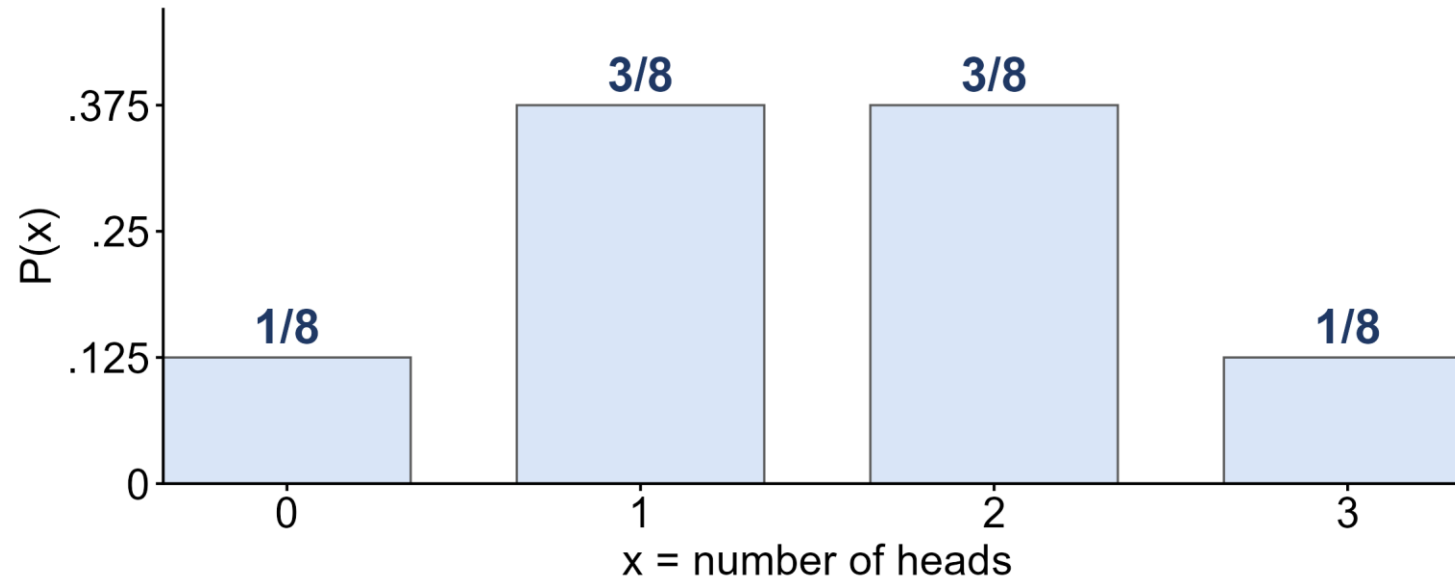
$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

$X = 0: 1 \quad X = 1: 3 \quad X = 2: 3 \quad X = 3: 1 \quad (\text{out of } 8)$

Discrete probability distributions

probability distribution

a function that assigns a probability to each possible value of X ; each is between 0 and 1, and they add to 1



$1 + 3 + 3 + 1 = 8$, so the probabilities add to $8/8 = 1$

Example: using a distribution

X = the number of heads in 3 tosses of a fair coin.

$$P(0) = 1/8 \quad P(1) = 3/8 \quad P(2) = 3/8 \quad P(3) = 1/8$$

(a) What is $P(X \geq 2)$?

(b) What is $P(X \leq 1)$?

(c) What is $P(1 \leq X \leq 2)$?

$$(a) \quad 3/8 + 1/8 = 4/8$$

$$(b) \quad 1 - P(X \geq 2) = 1 - 4/8 = 4/8$$

$$(c) \quad 3/8 + 3/8 = 6/8$$

The mean of a random variable

expected value μ

the long-run average of X : $\mu = \sum x \cdot P(x)$

variance σ^2

$\sigma^2 = \sum (x - \mu)^2 \cdot P(x)$, and the standard deviation is $\sigma = \sqrt{\sigma^2}$

three coin tosses

$\mu = 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = 1.5$ heads

Example: grade points

A STAT 251 exam grade on the 4-point scale, for a student chosen at random:

x	0	1	2	3	4
grade	F	D	C	B	A
P(x)	0.05	0.10	0.30	0.25	0.30

(a) What is $P(\text{better than a C})$?

(b) What is the mean grade, μ ?

$$\text{(a) } P(X = 3) + P(X = 4) = 0.25 + 0.30 = 0.55$$

$$\text{(b) } \mu = 0(0.05) + 1(0.10) + 2(0.30) + 3(0.25) + 4(0.30) = 2.65$$

What you should know: probability

probability

long-run relative frequency, between 0 and 1; the law of large numbers is why it works

the rules

$P(A') = 1 - P(A)$; $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; independent: $P(A \cap B) = P(A) P(B)$

random variables

a discrete distribution lists the values of X and probabilities that add to 1; $\mu = \sum x P(x)$